

Wrist Complex

→ They are 8 Carpal bones

Scaphoid

Lunate

Triquetrum

Pisiform

Trapezium

Trapezoid

Capitate

Hamate

[She Look Too Pretty]
(proximal bones)

[Try To Catch Her]
(Distal bones)

→ Wrist complex consist of 2 jts

RadioCarpal jts

Midcarpal jts

} Two jt System

- ↑ ROM

- ↓ changes of structural pinch

- Can hold ↑ pressure

Function -

→ Controls length - tension relationship

→ Anatomy varies which can change how function occurs.

Motion -

→ It is a Bi-Axial Jt

Flexion :- $0 - 65/85^\circ$

Extension :- $0 - 60/85^\circ$

Radial deviation :- $0 - 15/20^\circ$

Ulnar deviation :- $0 - 20/45^\circ$

Radio Carpal It :-

Proximal Radio Carpal jt / Articulation -

→ Articulation with Radius and Radio-Ulnar disc (TFCC) (Triangular fibrocartilage complex).

→ It has bi-Concave Curvature - long, shallow.

→ Lateral Radial facet - Scaphoid

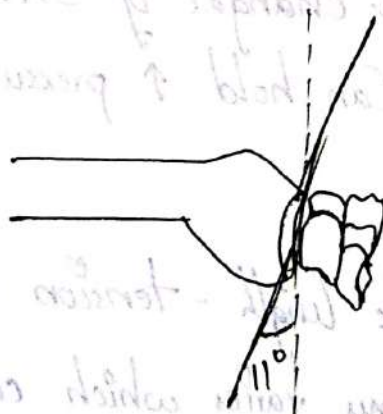
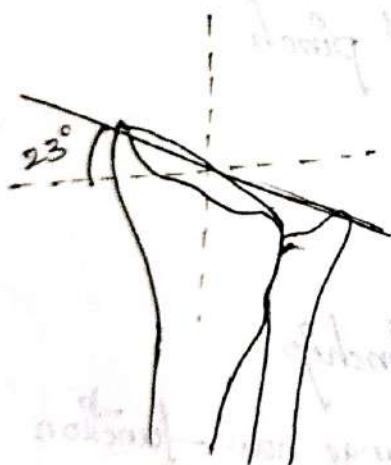
Medial Radial facet - Lunate

TFCC - Triquetrum and Lunate

→ 2 angles formed at the jt (^{Avg} Angular Inclination)

(1) Average inclination of Radius (23°)

(2) Ulnar tilt (11°)



Distal Radio-^{Carpal} Ulnar Articulation / It -

→ It articulates with Scaphoid, Lunate, Triquetrum and pisiform, here pisiform has 1 movement arm of Flexor Carpi Ulnar.

→ It is a Bi-Convex, can change shape.

Extensor Carpi Ulnae Sheath.

Meniscus Homologue

Ulnolunate / Ulno-Triquetral Ligament

Volar / Dorsal Radioulnar Ligament

→ ECU tendon → Connected to Lamina of TFCC
through ECU sheath and triquetrum, Hammae, Base
of 5th MCP.

→ Irregular Connective tissue - Meniscus Homologue

→ Overall articular incongruence exists - Contact is
not more than 40% at one time.

Radio Ulnar Variance :-

Positive Ulnar variance -

→ Impingement of TFCC b/w Ulna and Triquetrum.

→ Thinner TFCC.

→ Long Ulna after distal radius fracture.

→ Pain might be present on Pronation and Ulnar deviat?

Treatment :- If leveling [Ulnar shortening]

Negative Ulnar Variance :-

→ Relatively Short Ulna. Abnormal force distribution.

→ Thicker TFCC

→ Degeneration of Radio-Carpal jt.

→ Avascular Necrosis of lunate - Kienbock's disease.

Treatment :- Lengthening Ulna, Shortening Radius and
Fusing Carpal.

Mid Carpal Jt :-

→ Scaphoid, Lunate, Triquetrum articulation with Trapezium, Trapezoid, Capitate and Hamate.

Functional Unit :-

→ Anatomically separate from Radio-Carpal jt.
→ Capsule and Synovial lining is continuous with Carpals and Carpo-metacarpal jts.

→ Reciprocally Concave - Convex.

Union → Equally force distribution.

→ Foundation of transverse arches and longitudinal arches.

Ligaments -

Individual different exists in structure of carpus & ligaments.

→ Articular Stability guides and checks motion b/w Carpal bones.

Extrinsic -

→ Protects extrinsic slow healing intrinsic ligament, by taking forces first.

Intrinsic -

→ Strong, less stiff intrinsic rely on synovial fluid for nutrition.

* Volar ligaments are much more thicker than dorsal ligaments.

Extrinsic ligaments - Volar ligaments

Radio Carpal ligaments -

- 1) Radio Scapho Capitate (Radio Capitate)
- 2) Radio lunate Triquetral (Radio lunate)
- 3) Radio Scapho lunate - Conduit for Neurovascu-
larity to Scapho lunate Joint.
- 4) Radio Collateral ligament.

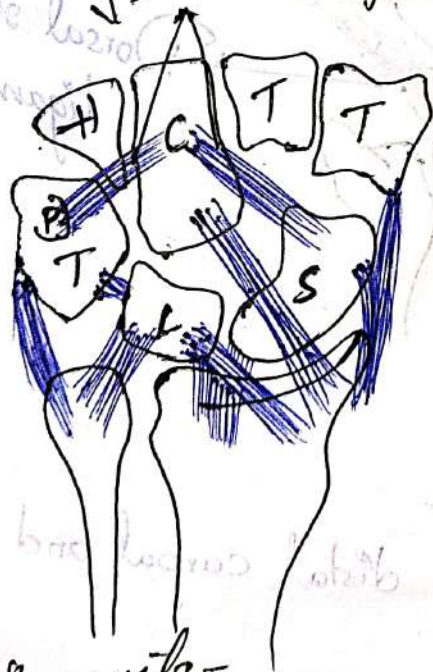
Ulnocarpal ligaments -

→ Complex - TFCC - Disc - Meniscus Homologue.

- 1) Ulnolunate Ligament

- 2) Ulnar Collateral ligament

↓ Deltoid ligament



Intrinsic ligaments -

- 1) Scapholunate - Scaphoid stability

↓
Arasclae - Degeneration.

2) Lunotriquetral - Lunatic Stability.
Instability might be seen with concomitant injury to extrinsic ligaments.

Dorsal ligaments -

- 1) Dorsal Radio-Carpal ligament
Oblique ligament - Helps offset sliding of proximal "carpal condyle" on Radius.
- 2) Dorsal Inter Carpal ligament.
Horizontal ligament [Both together form 'V' shaped stability of Radio-Carpal]



Kinematics :-

Flexion → Extension -
→ Active extension at distal carpal and MCP till neutral.

→ At Neutral Scapho-Capitate ligaments puts Scaphoid and Capitate in close pack position.

- Neutral → 45° Scaphoid and distal carpal move on Relatively fixed lunate and Triquetrum.
- At 45° Scapho-lunate Ligament puts Scaphoid and lunate in close pack position.
- This unites all carpals and they function as a unit.
- Complete extension occurs when proximal articulating surface of carpals move as a unit on radius and TFCC with all ligament taut.

Vice-Versa for Extension → Flexion.

- Inter carpal motion (proximal) Vary both in magnitude and direction with Axial load and Flexion/Extension/Radial/Ulnar deviation.

Center of Rotation :-

- Head of the capitate (Keystone).
- Center of fixed Carpal Arch Both A-P and Coronal Axis pass through capitate (flexion-extension) (Radial & Ulnar deviation).
- Computer modeling and cadaver study tells ↑ inter articular contact pattern with extension.

ROM -

Flexion - $(0-90^\circ)$

Extension - $(0-80^\circ)$

Functional ranges

54°

60°

Radial and Ulnar Deviation-

- Proximal Carpal bones display [Reciprocal Motion].

Radial Deviation-

- Distal carpal moves along with M.C.
- Proximal carpal slide ulnarily on Radius.
- Proximal carpal also go for flexion.
- Distal carpal go for Extension.
- Magnitude of motion can vary based on ligament laxity.

Opposite motion happens in Ulnar deviation.

ROM -

Functional range

Radial deviation ($0-20^{\circ}$)

17°

Ulnar deviation ($0-40^{\circ}$)

40°

- ROM is $\uparrow$ when wrist is Neutral.
- When Extended $\rightarrow$ carpal bones goes into close pack position with all locked $\rightarrow$ $\downarrow$ motion in Radial and Ulnar deviation.
- When Flexed $\rightarrow$ Carpal bones goes into loose pack position with all loosed $\rightarrow$ No movement occur.
- Extension + Ulnar deviation has maximum Scaphoid-lunate contact.
(20°) + (10°) $\rightarrow$ Stable Base + Optimal functional position.

Wrist Instability :-

Dorsal Intercalated Segment Instability (DISI) -

→ Scapho-lunate ligament injury removes synergistic stabilization.

→ Scaphoid collapses into flexion - (Subluxes) in dynamic and static.

→ Lunate and triquetrum → Extended → follow natural tendency all act like unconstrained segment

→ Carpals move into flexion.

→ Zigzag pattern of all 3 segments

→ Contact pressure b/w radius and scaphoid ↑
Capitate can migrate b/w scaphoid and lunate →

Degeneration.

→ Scapho-lunate Advanced Collapses (SLAC) in long term.

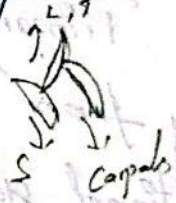
Volar Intercalated Segment Instability (VISI) -

→ Lunate - Luno-Triquetral ligament injury.

→ Lunate and scaphoid fall into flexion and triquetrum is unable to join lunate to counter balance the movement of scaphoid.

→ Ulnar - Peri-lunate Instability.

→ Triquetrum extends with other distal carpal hence proximal carpal stability is imp for wrist function.



Flexors or Volar Wrist Muscles :-

length-tension relationship stability and force.

→ Flexor Carpi Radialis -

Primary action is flexion at wrist works with extensor carpi radialis longus or offsets the extension produced by the muscle.

→ Flexor Carpi Ulnaris -

Primary action - flexion of wrist. Tendon placed away from the axis → ↑ force and movement → Functional relevance → Chopping wood.

→ Flexor Digitorum Profundus -

Flexors of fingers active/passive insufficiency longer and deeper → Crosses more joints.

→ Flexor Digitorum Superficialis -

Flexor of fingers but more of wrist flexors.

Multifid muscle → Extensors stabilize, action can vary with Radial/Ulnar deviation.

→ Flexor Pollicis Longus -

Flexor of Thumb and also stabilizes radial deviation.

→ Palmaris Longus -

Primary action - flexion

Absent Ulnar/Radial in 14% population → Sacrificed for Surgery.

Dorsal Wrist Muscles / Musculature :-

→ 9 muscles,

Three are primary muscles of wrist muscles

① Extensor Carpi radialis longus

② Extensor Carpi Radialis Brevis

③ Extensor Carpi Ulnaris.

Six are Secondary muscles of fingers & thumb muscles

① Extensor digitorum Communis [Extensor digitorum]

② Extensor indicis proprius [Extensor indicis].

③ Extensor digiti minimi

④ Extensor pollicis longus

⑤ Extensor pollicis brevis

⑥ Abductor pollicis longus.

Hand Complex

→ Hand consists of 5 digits : 4 fingers
1 Thumb

→ Each digit has a Carpometacarpal jt & a metacarpophalangeal jt.

→ The fingers each have 2 interphalangeal (IP) jts.

- Proximal interphalangeal (PIP)

- Distal interphalangeal (DIP)

Carpometacarpal jts of the fingers :-

→ Composed of the articulations b/w the distal carpal row & the bases of 2nd to 5th metacarpal jts.

→ 2nd metacarpal articulates

primarily - trapezoid

Secondarily - trapezium & Capitate.

→ 3rd metacarpal articulates with Capitate.

→ 4th metacarpal - Capitate & Hamate

→ 5th metacarpal - Hamate.

→ Carpometacarpal jts are supported by strong transverse & weak longitudinal ligaments volarly & dorsally.

→ Deep transverse metacarpal ligament spans the head of the 2nd through 4th metacarpals volarly.

→ Proximal transverse arch formed by the trapezoid, trapezium, Capitate, hamate.

→ Ligaments that maintain the arch are
→ Transverse Carpal ligament

→ Transversely oriented intercarpal ligaments.

ROM of Carpometacarpal jt :-

→ 2nd to 4th Carpometacarpal jt → plane synovial jt

- 1° - freedom.

- Flexion / Extension

→ 2nd & 3rd Carpometacarpal jts are immobile

- 0° of freedom.

→ 4th Carpometacarpal jt has flexion / Extension

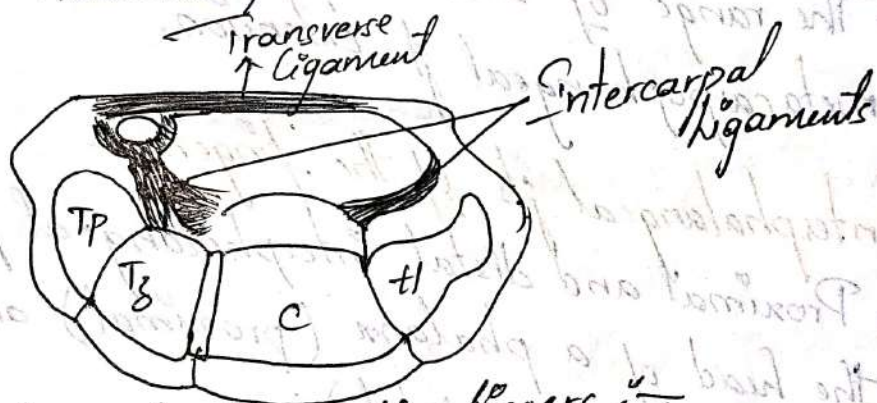
→ 5th Carpometacarpal jt is a saddle jt

- 2° of freedom

- Flexion / Extension

- Abduction / Adduction

→



Metacarpophalangeal jts of the fingers :-

→ Articulation b/w Convex metacarpal head proximally and concave metacarpal base distally.

→ It is the condyloid type of jt with 2° freedom

Flexion / Extension

Abduction / Adduction.

→ It surrounded by a capsule that is lax in extension.

→ 2 Collateral ligaments volarly located deep transverse metacarpal ligament enhance jt stability.

Ligaments :-

→ Radial & Ulnar collateral ligaments of metacarpophalangeal jt composed of 2 parts.

- 1) Collateral ligament proper
- 2) Accessory Collateral ligament

ROM :-

→ Flexion/Extension increases radially and ulnarly, with index finger - 90° flexion

little finger - 110° flexion.

→ Hyperextension varies among individuals based on their flexibility.

→ The range of abduction/adduction is maximal in metacarpophalangeal jt extension.

Interphalangeal jts of the fingers :-

→ Proximal and distal interphalangeal jts composed of the head of a phalanx (proximal) and the base of the phalanx dig (distal).

→ They are true synovial hinge jt

1° of freedom

Flexion/Extension.

→ Total range of flexion and extension available to the index finger is greater at the proximal interphalangeal jt (100° to 110°), distal interphalangeal jt (80°).

Extrinsic finger flexors :-

→ Extrinsic muscles - Proximal attachments (muscles of fingers and thumb) of the wrist.

→ Intrinsic muscle - Distal attachments of wrist.

→ Extrinsic Muscles

→ Flexors
→ Extensors

→ Extrinsic muscles (flexion of fingers)

1) Flexor digitorum superficialis.

2) Flexor digitorum profundus.

Flexor digitorum superficialis -

Flexes the proximal interphalangeal jt also contributes to metacarpophalangeal jt flexion.

Flexor digitorum profundus -

Flex the metacarpophalangeal, proximal interphalangeal & distal interphalangeal joints.

Mechanism of finger flexion :-

→ Flexor digitorum superficialis & flexor digitorum profundus ~~are~~ helps in flexor gliding mechanism.

→ Gliding mechanism consists of the flexor retinacula, bursae and digital tendon sheaths.

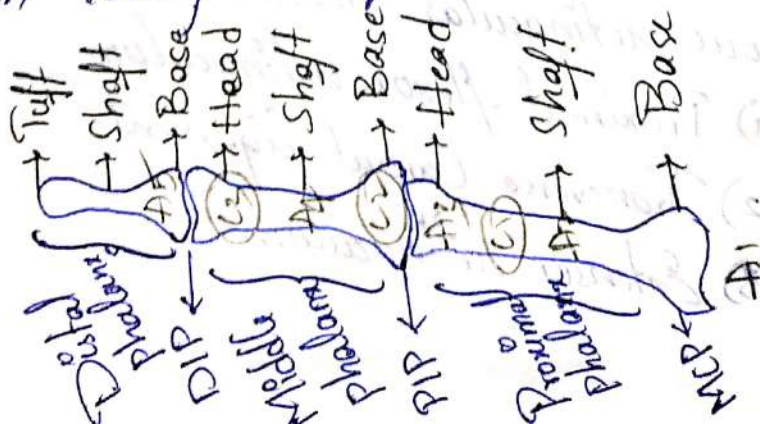
→ Fibrous retinacular structure consists of

1) Proximal flexor retinaculum

2) Transverse Carpal Ligament

3) Extensor retinaculum

- The flexor sublimaris tethers the long flexor tendon to the hand.
- The bursae & tendon sheaths facilitate friction free excursion of the tendons on fibrous sublimaris.
- The tendons of 2 muscles pass through fibrous tunnel that comprised of 5 transversely oriented annular pulleys & 3 cruciate pulleys.
- A1 pulley : lies at the head of metacarpal.
- A2 pulley : Along the volar midshaft of proximal phalanx.
- A3 pulley : lies at distal most part of the proximal phalanx.
- A4 pulley : lies centrally on the middle phalanx.
- A5 pulley : lie at the base of the distal phalanx.
- C₁ pulley : located b/w A₂ and A₃ pulleys
- C₂ pulley : lies b/w A₃ and A₄ pulleys
- C₃ pulley : lies b/w A₄ and A₅ pulleys.
- Function of annular pulleys.
 - 1) Keep the flexor tendons close to the bone.
 - 2) Allowing only min amount of bowstringing
 - 3) Migration volarly from jt axis.



Extrinsic finger Extensors :-

→ The extrinsic finger extensors are the

1) Extensor digitorum communis

2) Extensor indicis proprius

3) Extensor digiti minimi muscle.

→ Pass from the forearm to the hand beneath the extensor retinaculum.

→ The extensor digitorum communis tendons of one finger may also be connected to the tendon or tendons of an adjacent finger by junctura tendinae.

Extensor mechanism :-

→ It is formed by

1) The tendons of extensor digitorum communis muscle (with extensor indicis proprius & extensor digiti minimi).

2) Extensor hood

3) The central tendon

4) Lateral bands are merge into terminal tendon.

→ The lateral bands are interconnected dorsally by a triangular bands of superficial fibers known as dorsal retinacular ligaments.

→ The dorsal interossei (OI), volar interossei (VI) & lumbrical muscles are the active component of the extensor mechanism.

Structure of the thumb :-

Carpometacarpal jt of the thumb :-

→ Carpometacarpal / Trapeziummetacarpal jt of the thumb is the articulation b/w the trapezium & the base of 1st metacarpal.

→ It is a saddle type jt
2° freedom

Flexion / Extension

Abduction / Adduction

Circumduction (opposition)

→ Saddle shaped portion of trapezium is concave in sagittal plane & convex in frontal plane.

→ The base of the 1st metacarpal has a reciprocal shape to that of trapezium.

→ Opposition is sequentially, abduction, flexion & adduction of the 1st metacarpal.

Extrinsic Thumb Muscles :-

1) Flexor pollicis longus

2) Extensor pollicis brevis

3) Extensor pollicis longus

4) Abductor pollicis longus.

Intrinsic Thumb Muscles :-

→ 5 thenar muscles

1) Opponens pollicis

2) Abductor pollicis brevis

- 3) Flexor pollicis brevis
- 4) Adductor pollicis
- 5) 1st volar interossei muscles.

Perhension :-

- Perhension activities of hand involve the grasping or taking, hold of an object b/w any 2 surfaces in the hand.
- Thumb participates in most but not all the perhension tasks.
- Perhension can be categorized or either
 - 1) power grip (full hand perhension)
 - 2) Precision handling (finger-Thumb perhension)

Power Grip :-

- The fingers in power grip usually function in concert to clamp on & hold on object into the palm.
- 4 types of power grip.
 - 1) Cylindrical grip
 - 2) Spherical grip
 - 3) Hook grip
 - 4) Lateral prehension

① Cylindrical Grip :-

- It involves the use of flexors to position the fingers around & maintain grasp on an object.
- Muscles of hypothenar eminence usually are active in cylindrical grip.

→ The thumb usually comes around the object, then flexes & adducts to close the vise.
→ wrist is in neutral flexion/extension & slight ulnar deviation.

→ The ring & little finger seem to serve as weaker but more mobile fingers assists to the more stable & stronger index & middle fingers.

Ex:- Turning a door knob.

② Spherical grip :-

→ Spherical grip is similar in most respects to cylindrical grip.

→ Greater spread of the fingers to encompass the object.

→ Opening the hand during object approach & object release is primarily on extensor function, calling in activity of lumbricals.

Ex:- Holding a ball.

③ Hook Grip :-

→ Specialized form of prehension.

→ It is a function primarily of the fingers.

→ It may include the palm but never includes the thumb.

→ Sustained for prolonged periods of time.

→ Major muscular activity is provided by flexor digitorum profundus & flexor digitorum superficialis muscles.

Ex: ~~Gallin~~
Carrying a briefcase or books at his side.

Lateral prehension :-

- It is a unique form of grasp.
- Contact occurs b/w 2 adjacent fingers.
- The metacarpophalangeal & interphalangeal jts are in extension.
- Metacarpophalangeal joints are simultaneously abduct & adduct.

→ Example: Holding a card b/w 2 fingers.

Precision Handling :-

- Require much finer motor control, & they are more dependent on intact sensation.
- The thumb is generally abducted & rotated from the palm.

→ 3 types of precision handling

- 1) Pad to Pad precision
- 2) Tip to tip precision
- 3) Pad to side precision

① Pad to Pad Precision :-

- Involves opposition of the pad or pulp of the thumb to the pad of the finger.
- Metacarpophalangeal & proximal interphalangeal jts of fingers are partially flexed.
- Distal interphalangeal jt may be fully extended.

→ Thumb is held in opposition.

② Tip - to tip prehension :-

- The interphalangeal jts of the finger & thumb must have the range & available muscle force to create nearly fully jts flexion.
- Metacarpophalangeal joints of opposing fingers must be ulnarly deviated.
- If active force of interphalangeal flexion & metacarpophalangeal joint ulnar deviation can't be provided tip to tip prehension cannot be performed effectively. [Holding a pen].

③ Pad - to ^{Side} Pad prehension :-

- Also known as key grip or lateral pinch.
- Because key is held b/w the pad of thumb & side of the index finger.
- Thumb is more adducted & less rotated.
- Activity of flexor pollicis brevis increases & that of opponens pollicis decreases.

* Tenodesis :-

- The phenomenon of using active wrist extension to passively close the fingers & passive wrist flexion to passively open the fingers is known as Tenodesis.